Emotional Support Dialogue Generation: An In-context Learning Approach

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August 17, 2026

Abstract

Rapid advances in large language models (LLMs) create new opportunities for Emotional Support Conversation (ESC), a task that is becoming increasingly relevant as people face changing social and psychological challenges. However frequent model updates and the cost of task-specific fine-tuning make conventional deployment increasingly difficult to sustain. This study investigates whether in-context learning (ICL) can provide a lightweight and adaptable alternative for ESC generation. Meanwhile, the expanding general capabilities of LLMs call into question whether task-specific parameter updates remain necessary for every downstream task. This study therefore investigates the feasibility of shifting ESC from the fine-tuning paradigm toward In-Context Learning (ICL).

We explore three representative ICL paradigms, namely Chain-of-Thought (CoT), Multi-Agent Systems (MAS), and Skill Distillation, by developing corresponding models, and compare them with fine-tuned and prompting-based baselines. We also examine whether functional innovations previously developed in fine-tuning-based ESC research can be explicitly transferred into prompts. The results demonstrate that carefully designed ICL methods can outperform the evaluated fine-tuned baselines with a comparable backbone, confirming that insights from earlier ESC research can remain effective when reformulated as prompting mechanisms. We also discuss that, while limiting its suitability for lightweight applications, greater architectural and reasoning complexity does not necessarily produce proportional gains.

The distilled supporter-skill presents the most striking result. By distilling task-specific behavioral knowledge from dialogue data into a compact natural-language prompt, it achieves the strongest overall performance while retaining near one-shot inference efficiency. This finding points toward a possible methodological shift from manually constructing increasingly elaborate reasoning pipelines to enabling LLMs to discover, organize, and refine task-specific operating principles through meta-prompting and iterative prompting. Analogous to the transition from handcrafted features to

learned representations in machine learning, data-driven prompt and skill discovery may offer a more flexible and scalable direction for future ESC systems.

The code of this study can be found in GitHub repository.[1]

1 Introduction

1.1 Background

The rapid evolution of Large Language Models (LLMs) has demonstrated remarkable potential in emotional and context-aware intelligence, transitioning these systems from mere task-oriented tools to agents with empathetic abilities. Parallel to this technical progression, the social demand for conversational agents (CAs) capable of providing proactive psychological support arises rapidly. Recent literature proves simultaneously the efficacy of chat-bots in mitigating emotional distress and raises ethical concerns regarding users' over-reliance and possible addiction (Yankouskaya et al., 2025). Consequently, the Emotional Support Conversation (ESC) task (Liu et al., 2021) has gained more attention in human-computer interaction (HCI), offering potential solutions across diverse scenarios from peer companionship (Liu et al., 2024a), psychotherapy (Chen et al., 2023; Liu et al., 2023; Srivastava et al., 2023) to empathetic customer service (Brun et al., 2025).

As LLMs continue to scale and expand from millions to hundreds of billions of parameters, the computational overhead, encompassing GPU hours and memory constraints, alongside the extensive data engineering required for full-parameter fine-tuning, has become increasingly prohibitive for small-to-medium enterprises (SMEs) engaged in rapid prototyping. Conversely, recent research highlights the "emergent abilities" of advanced models such as Llama-3 and GPT-4, which enable complex reasoning through In-Context Learning (ICL) paradigm. Techniques including Chain-of-Thought (CoT) prompting(Wei et al., 2022), multi-agent systems, and recent skill distillation(Anthropic, 2026; OpenAI, 2026) approaches deliver a lightweight, flexible alternative to conventional model deployment pipelines.

Currently, the methodological landscape (Deng et al., 2024) of ESC tasks remains predominantly grounded in Supervised Fine-Tuning (SFT), frequently augmented by Reinforcement Learning (RL) (Zhou et al., 2023; Srivastava et al., 2023) or the integration of external commonsense knowledge and auxiliary models (Tu et al., 2022). While a small body of recent studies has explored CoT reasoning and multi-agent system for ESC (Li et al., 2024b; Xu et al., 2025), there remains a lack of research that systematically examines whether the methodological and architectural innovations previously proposed for fine-tuning-based ESC models can be transferred to the prompt engineering paradigm.

[1] `https://agentskills.io/home`

1.2 Research Questions

Main Research Question: How effective is In-context Learning serving as a lightweight and adaptive alternative to fine-tuned Emotional Support Dialogue Systems?

RQ1: How does an advanced moderate-scale LLM (e.g., Llama-3-8B) with multi-aspect prompt engineering assistant performs on the ESConv benchmark, in comparison to fine-tuned baselines?

RQ2: To what extent can prompt engineering replicate and benefit from architectural innovations in earlier ESC research (typically achieved through Fine-tuning, Attention-based feature extraction, Reinforcement Learning and External Commonsense Knowledge), for example through CoT reasoning, context management, user profiling, emotion recognition and causal inference, and support strategies selection?

RQ3: How do CoT prompting and role-assigned multi-agent frameworks perform against each other when equipped with essentially identical base prompts?

RQ4: Can prompts automatically summarized via skill distillation deliver comparable performance to manually crafted prompt templates?

1.3 Purpose & Goals

For this reason, the present study systematically investigates the feasibility of migrating existing fine-tuning-based methodologies to ICL paradigm, integrateing the typical improvement directions in this field within the CoT and multi-agent framework respectfully. These directions include the selection and adherence to supportive dialogue strategies, user emotion recognition, emotional cause analysis, and self-reflection mechanisms designed to guide positive emotional and cognitive transitions. Also, this work examines application of recent skill-distillation techniques on ESC tasks.

1.4 Research Methodology

We validate the feasibility of applying ICL to the ESC task through comparative experiments against fine-tuned baseline models on Llama-3-8B, and further discuss the performance impacts of different prompt modules and techniques via comparative experiments.

The present research represents an intermediate step toward that long-term objective. Specifically, it aims to identify and address the technical limitations of hard prompt engineering relative to fine-tuning, with the goal of developing a functionally robust emotional support dialogue agent that can support subsequent stages of system development.

2 Related works

2.1 Proactive Conversational Agents

Proactive conversational agents (PCAs) have emerged as a transformative paradigm in conversational artificial intelligence, shifting the core design philosophy from passive, instruction-following systems to initiative-acting agents.

Deng et al. (2024) advance a three-dimensional taxonomy of PCAs, categorizing agents by their proficiency in Intelligence, Adaptivity, and Civility, and investigates their data preparation strategies and model learning paradigms for proactive conversational systems. Deng et al. (2025) defines PCAs as systems capable of proactive conversational planning, goal-directed dialogue steering (rather than passive compliance with user-proposed tasks), and anticipation and reflection on the impacts of conversational actions. This work proposes a granularized taxonomy of PCA tasks as well as main technical approaches in context of LLM.

Despite PCAs encompassing capabilities such as initiating sub-dialogues for clarification, mainstream research remains confined to in-conversation proactivity rather than the ability to proactively initiate dialogues themselves. Lu et al. (2024) pioneered a PCA capable of proactively proposing task requests without explicit user instructions, leveraging environmental observations of user activities and system states to anticipate latent needs, and constructed a dataset covering various work scenarios. Liu et al. (2025) introduced an inner thought mechanism enabled by additional working memory, allowing agents to simulate speaking motivations and assess optimal interjection timing in group conversations. Liu et al. (2024a) developed ComPeer, a peer support agent which boosts user engagement and stress relief through adaptive proactive messages.

2.2 Emotion Supportive Conversation Generation

Emotional Support Conversation (ESC) is a core representative task in the field of PCAs, focusing on alleviating users' negative emotions and promoting their psychological well-being. Liu et al. (2021) pioneered the task of emotional support conversation generation, formally defined ESC and constructed the ESConv dataset, an corpus of supportive dialogues between help-seekers and trained supporters, annotated with detailed labels for eight support strategies, enabling the training and evaluation of empathetic response generation systems. This work established a foundational benchmark for ESC research.

2.2.1 Fine-tuning (and RL) Approach

Liu et al. (2021) proposed that Emotional Support Conversation systems should explicitly select and follow dialogue strategies grounded in Hill's Helping Skills Theory, which has since become standard practice in the field. Tu et al. (2022) incorporated the pretrained commonsense model COMET to assist fine-grained emotion understanding. It further modeled support strategies as a probability distribution over a strategy codebook, enabling mixed-strategy-guided response generation. Inspired by the A* search algorithm, Cheng et al. (2022) introduced lookahead heuristics, and modeled emotional intensity, emotion causes, and persuasive effects. This approach estimates long-term user feedback to optimize strategy selection. In the similar line, Zhou et al. (2023) introduced reinforcement learning (RL) with a mixture-of-experts (MoE) architecture to incentivize models to guide users toward positive emotional transitions rather than merely imitating human dialogue patterns. Cheng et al. (2023) propose the PAL model that first dynamically compresses seekers' persona information from conversation history, and then integrates this persona with a strategy-based generation method to produce personalized supportive responses. Inspired by these practices, this paper aims to integrate these mechanisms into a unified ICL framework under CoT-based reasoning and multi-agent systems.

Memory management, particularly issues such as external variable (EV) caching, has emerged as a significant challenge for large language model (LLM) inference in long-context tasks. (Shi et al., 2026) In relevant contexts, existing studies highlight that the precise selection and management of context are crucial for the quality of ESC generation. Peng et al. (2022) proposed a global-to-local graph network to models hierarchical relationships between global causes (dialogue-level problematic cause), local intentions (turn-level mental states), and dialogue history through a graph reasoner. Peng et al. (2023) further improved context management by a double control reader that constructs and restricts bidirectional flows (context-to-strategy and strategy-to-context), dynamically focusing on strategy-related context in long conversations. The approach effectively mitigates the risk of context attenuation in long interactions and improves the consistency between selected supportive strategies generated responses.

Among works of mental health counseling scenarios, Chen et al. (2023) constructed a large-scale augmented Chinese dataset combining crowdsourced single-turn counseling data and ChatGPT-driven rewriting under empathy-constrained prompts. Fine-tuning on this dataset addresses the over-suggestion inclination of generic LLMs. Liu et al. (2023) applied fine-tuned LLMs to professional psychological counseling, and proposed a GPT-4-powered evaluation framework with seven counseling-specific metrics. Srivastava et al. (2023) leveraged transformer-reinforcement-learning (TRL) with Proximal Policy Optimization (PPO) to improve counseling strategies, proposing the READER model that jointly predicts response-acts and generates appropriate and semantically rich responses.

2.2.2 In-Context Learning Approach

Chain of Thoughts (CoT)

Li et al. (2024b) proposed a plug-and-play prompting method based on CoT reasoning (Wei et al., 2022), namely Emotional CoT (ECoT), which integrates Goleman's emotional intelligence theory to guide LLMs through step-by-step emotional reasoning before responcing, including understanding context, recognizing others' emotions, recognizing self-emotions, managing self-emotions, and influencing others' emotions. The experiments indicated better alignment with human preference.

Multi Agent Systems (MAS)

Lai et al. (2025) extend the research scope of PCAs to multi-agent LLM collaborative scenarios, investigates the collaboration strategies of current designs, and empirically evaluates the performance of these strategies across three representative AC tasks. The work further finds multi-agent systems (MAS) underperform other enhancement strategies in emotion analysis tasks, yet they help capture dynamic emotional nuances and provide professional advice in ESC tasks. However, the MAS methods in this work are limited to parallel agent ensemble voting and conversational role-playing, which differ from mainstream general MAS designs, e.g. task decomposition and sub-task assignment.

Cheng et al. (2024) assigns different roles to agents, such as Explorer, Comforter and Advisor. Each agent follows an independent analysis process, including state tracking, progression and aspect-specific analysis. The agents then propose potential topics for the next round of dialogue, and the final topic is selected from these candidates based on top-k scoring. Therefore, to normalize the estimated progress from different aspect-specific agents and score the candidate topic list, additional training of a progress estimation network and a scoring head network is still required. In contrast, Xu et al. (2025) separates role assignment and expert collaboration into two stages. In the first stage, agents sequentially identify the user's emotion, causal events, and intentions. In the second stage, experts initialized with different strategy-response pairs engage in a discussion and vote to select the final response.

Role-playing

role-playing has become an important approach for leveraging the generalized language capabilities of LLMs to support further fine-tuning in various aspects. Li et al. (2023), Chen et al. (2023), and Li et al. (2024a) utilized LLMs to simulate emotional support seekers and supporters to generate or complete dialogues, thereby constructing multi-turn dialogue datasets.

Wu et al. (2025) leveraged standard psychological inventories to summarize conversationalists' personality traits, demonstrating LLMs' effectiveness in simulating specific personas. The study also noted that capturing help-seekers' personality characteristics is crucial for delivering reasonable and effective emotional support, as proven persona infusion modifies support strategy distribution and improves the personalization and empathy of agents. Tu et al. (2025) assigned specific personalities to dialogue participants.

Beyond dataset construction, several studies have also explored the direct application of role-playing strategies. For example, Liu et al. (2024a) and Brun et al. (2025) applied role-playing-based approaches to deploy LLMs in peer-support and customer service scenarios.

Limitations of hard-prompt approaches

Several works have identified limitations of hard prompt approaches. Cheng et al. (2023) point out that hard prompts typically constructed by simple concatenation of auxiliary information with dialogue history fail to achieve deep fusion between auxiliary information and context. Madani and Srihari (2025), and Mao et al. (2022) compare the performance of different predefined official prompt templates, as well as hard prompt paradigms against fine-tuning (FT) methods. They conclude out that hard prompts suffer from the "lost in the middle" effect in LLMs, where auxiliary instructions (containing strategy or persona information) tokens are overlooked in long interactions, especially when the prompt is positioned prior to the conversation history. Also, static prompts may contain outdated or incorrect information.

Drawing inspiration from related studies mentioned above in FT approach, we identify structured user modeling and context management as promising directions for addressing prompt degradation in future work. Maintaining explicit representations of user personality, emotional state, and ongoing discussion topics may mitigate prompt degradation and improve the robustness of prompt-based emotional support dialogue systems.

2.3 Skill Distillation (SD)

Skill distillation (SD) is a phenomenon-level emerging trend in 2026 that has attracted a lot of attention from AI practitioners and developers. It refers to extracting reusable skills from LLMs and turning them into explicit prompts or workflows.[2] Unlike traditional LLM distillation methods that align hidden representations, or reinforcement learning approaches that imitate action trajectories(Wang et al., 2026), skill distillation focuses on transferring higher-level skills and problem-solving patterns in a more explicit and human-interpretable way.

The COLLEAGUE.SKILL framework(Zhou et al., 2026) have fueled this trend forward, mainly due to its practical deployability and its provocative while controversial concept of "Digital Life" and of fixing workflow and capacity of former employee, or even behavior pattern of celebrities and ex-partners, for further use. In its latest update, the project has been renamed as dot-skill to emphasize its general applicability.

[2]For detailed definition and introduction on what agent skills are, see `https://agentskills.io/home`

The current implementation is constructed in a meta-prompting style, and provides three functional components: colleague-skill, relationship-skill, and colleague-skill. The primary objective of dot-skill is to reconstruct individual behavioral patterns through fine-grained, structured, and multi-level meta-prompting, and it decomposes personality into different aspect. Dor example, colleague-skill distinguishes between work capacity and persona, while relationship-skill explicitly separates an individual's actual behaviors from the user's perceptions and interpretations. Furthermore, relationship-skill reconstructs personas at multiple levels, including linguistic patterns in common expressions, emotional triggers, and conflict and repair patterns.

However, in the academic community, research on prompt-based skill distillation in the strict sense remains limited. Nevertheless, we notice that it shares important similarities with Reflexion(Shinn et al., 2023). Reflexion uses only prompting mechanisms to reflect on the relationship between action trajectories and negative rewards. It converts sparse rewards into natural-language feedback, which is then used as an updated prompt to improve future actor behavior. More recently, Badhe and Shah (2026) showed that LLM reasoning abilities can be distilled through prompting alone in a teacher-student framework, which appears to disclose standard workflows adopted by major tech enterprises.

Comparison with Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) constructs more reliable AI Expert Systems systems by integrating external resources such as databases, knowledge graphs and retrieval mechanisms. The primary goal of RAG is to reduce hallucination and improve factual consistency. In the field of emotional support dialogue, recent ICL approaches have explored RAG as a way to overcome the limitations of manually designed prompts.

Xu et al. (2025) retrieves similar conversations from the ESConv dataset to provide top-k relevant response strategies for further thought. Furthermore, Tu et al. (2025) proposes an incremental ICL framework by maintaining both LLM-simulated and real user interaction memories. After each conversation with a seeker, the framework extracts user personality and summarizes customized response tips through reflective questions. However, the paper does not reveal how personality information is embedded, or how reflection, memory retention and forgetting mechanisms are determined. Furthermore, analysis of its official GitHub implementation[3] indicates that prompt optimization is also used as an auxiliary.

2.4 Emotion Computation

Emotion Computation is an earlier and functionally constrained task related to ESC, primarily focuses on analyzing and recognizing user emotions. Liu et al. (2024b) introduced EmoLLMs, a series of emotion-focused large language models fine-tuned on the multi-task

[3] `https://github.com/HITSZ-HLT/Supportplay`

Affective Analysis Instruction Dataset (AAID), alongside Affective Evaluation Benchmark (AEB) covering classification and regression tasks.

Kim et al. (2025) proposed a modular prompt design framework for monitoring user emotions and mental health states.They decompose prompts into six core components (Persona, Task, N-shot Examples, Input, Output, Template) and systematically evaluates the impact of key components (Persona and Task Instruction) across four LLMs and five datasets, indicating the need and advantages of task-specific prompt design. However, Mao et al. (2022) discuss potential systematic biases when using pretrained LLMs for emotion analysis with hard prompts, including biases induced by task definition, word frequency and prompt and label-word forms, which undermines the reliability of hard prompt-based approaches in affective computing tasks.

Zhang et al. (2024) leveraged smartphone sensor data, including battery levels, screen unlocks, location information, application usage, keyboard usage, and communication traces, to infer user affective states via LLMs under zero-shot and few-shot settings. By converting numeric sensor features into textual descriptions, the study establishes a link between behavioral patterns and emotions, providing a valuable reference for integrating non-dialogue signals into ESC and emotion computing tasks.

3 Method

Multiple ICL approaches are implemented and examined, including Chain of Thoughts (CoT), Multi-agent System (MAS) and Skill Distillation (DS).

3.1 Module Design

Motivated by the representative studies under the fine-tuning paradigm reviewed in the related work (see 2.2.1), we identify and summarize several key design principles, motivations and reasoning underlying novel improvements and commonly adopted practices for ESC systems. Specifically, existing approaches commonly incorporate:

- Explicit modeling of support strategies, where predefined strategies are selected or combined to guide response generation.
- Fine-grained emotion recognition and causal analysis to understand the seeker's affective state and its underlying causes.
- Reflection on how supportive communication and persuasion can facilitate positive changes in the seeker's emotion and cognition.
- Extraction of the seeker's personality traits and preferences for personalized response generation.

- Hierarchical context modeling, where global dialogue context and fine-grained local context are separately summarized, with local contexts aligned with corresponding support strategies to improve response relevance and coherence while mitigating too generic responses and context degradation.

3.2 Chain of Thoughts

The ordering of reasoning modules is designed according to the dependency relationships among different types of information. Specifically, emotion recognition and context summarization provide fundamental information for subsequent reasoning, while the outputs of all other modules jointly contribute to strategy selection and final response generation process. Therefore, we construct the reasoning chain with the following sequential steps:

1. Identify the seeker's current emotional state and infer its causes.

2. Summarize both global context, including the initial motivation and overall background of seeking support, and local context, referring to the specific topic or event emphasized in the current dialogue turn.

3. Extract the seeker's personality characteristics, values, and interpersonal patterns.

4. Reflect on how supportive communication can facilitate positive emotional and cognitive transitions, followed by selecting the top-k applicable support strategies.

5. Analyze which aspects of the seeker's concerns can be addressed by each selected strategy, associate strategies with corresponding contextual information, and generate candidate responses.

6. Evaluate the generated candidates and synthesize the most appropriate response by selecting or integrating effective elements from different candidates.

Step (5) is applies to exploit strategy fusion and improve contextual alignment. The detailed prompt design is provided in Appendix A.1.

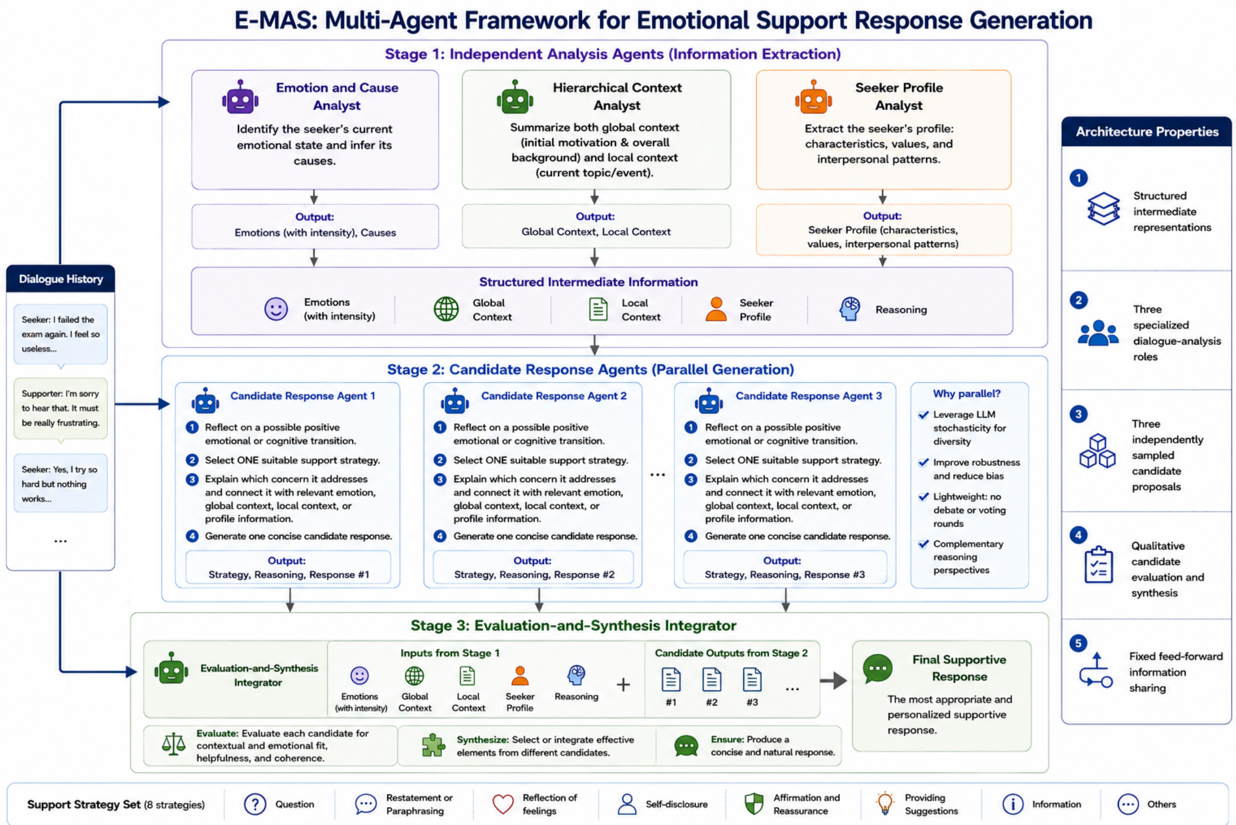

Figure 1: Overview of E-MAS.

3.3 Multi-agent Collaboration

The primary advantage of MAS lies in its ability to assign specialized roles and responsibilities to individual agents, enabling agents to focus on specific sub-tasks or adopt complementary differential perspectives. Furthermore, MAS provides an explicit mechanism for coordinating information exchange among agents, extending beyond conventional sequential generation paradigms.

In our framework, emotion recognition with causal inference, hierarchical context summarization, and user personality extraction are relatively independent subtasks. Therefore, assigning these components to separate agents enables independent intermediate reasoning, reduces the dependency among intermediate outputs, and introduces diverse perspectives. The information produced by these agents is subsequently integrated during the strategy planning and response refinement stages.

Multi-agent prompting with multiple independent agents sharing identical instructions is also a widely adopted strategy for exploiting the stochasticity and generalization capability of LLMs. Multiple generations can improve output diversity and robustness at the same time. However, for ESC scenarios, existing approaches based on parallel agents(Xu et al., 2025) often introduce multi-round debates and voting mechanisms for selecting candidate topics or strategies. In our setting, we argue that such mechanisms may introduce unnecessary computational overhead. Specifically, a substantial amount of inference budget is allocated to strategy selection, while additional engineering complexity is introduced in maintaining topic consistency, assigning initial agent positions, and controlling interaction rounds. These mechanisms are also less aligned with the heuristic role of strategy selection within the overall response generation pipeline.

Moreover, COOPER (Cheng et al., 2024) assigns multiple (m=4) candidate strategies to multiple (k=3) individual agents. However, ESC datasets generally adopt a relatively limited set of predefined support strategies, typically containing only 8–15 categories (Madani and Srihari, 2025). Therefore, instead of employing inter-agent debate and voting mechanisms, our framework adopts three independent strategy-generation agents (k=3), where each agent selects suitable strategies and generates candidate responses independently. A final integration agent then aggregates the intermediate outputs and produces the final refined response.

Figure 1 illustrates the overall E-MAS pipeline. The intermediate outputs and formatting details are simplified for readability; the complete specifications can be found in the prompts of the E-MAS agents in Appendix A.2.

3.4 Skill Distillation

To investigate the effectiveness of SD in ESC scenarios, we utilize the official GitHub repository of dot-skill[4] to extract the persona of the emotional supporter in the ESConv dataset. However, it should be noted that dot-skill is originally designed for reconstructing the personality, behavioral patterns, and user-specific connections of particular individuals, with particular emphasis on the Memory Signature between the reconstructed persona and the user. In contrast, ESConv conversations are collected from diverse sources, and our setting does not require maintaining memories between the supporter and seeker.

To adapt dot-skill to our experimental setting, we use relationship.skill rather than colleague.skill for distillation. The latter is originally written in Chinese, contains job-and-tech-stack-related prompts that are highly specific to the corporate setting, and assumes predefined personality and behavioral patterns for a workplace colleague, which do not fit the ESC task.

We remove content related to memory signatures and specific conversation histories from the skill configuration .md files. We also explicitly instruct the dot-skill builder to reconstruct a general supportive conversational agent, rather than a memory-dependent individual identity.

We use GPT-5.6 Solar as the host model and provide the full ESConv training set as

[4] `https://github.com/titanwings/colleague-skill`

resource. The generated skill.md is limited to 2,000 characters. The detailed instructions and the distilled skill.md are provided in Appendix A.4.

4 Experiments

4.1 LLM-based Evaluation Metrics

Among works validated on the ESConv dataset, the core automatic evaluation metrics include perplexity (PPL), BLEU-2 (B2) (Papineni et al., 2002), and ROUGE-L (R-L) (Lin, 2004). However, conventional automatic metrics fail to capture affective and supportive qualities, as they only measure lexical or semantic overlap instead of assessing empathy and supportiveness. To mitigate this limitation, user studies serve for achieving subjective evaluation, while LLM-based evaluation has emerged as a cost-effective and scalable alternative when large-scale user experiments are infeasible and when evaluation criteria are complex.

Li et al. (2024b) proposed the Emotional Generation Score (EGS), which integrates Goleman's Emotional Intelligence Theory. EGS evaluates generated responses across dimensions including relevance, avoidance of negative attitudes, reflection of positive emotions, and facilitation of positive emotional transitions. Using GPT-3.5 as the discriminative model, they verified that EGS is consistent with human expert ratings. Madani and Srihari (2025) adopted GPT-4o as the judge LLM to assess how well conversation systems follow given support strategies. Wu et al. (2025) employed GPT-4o-mini, Claude-3.5-Haiku, and Llama-3.1-8B-Instruct to identify and simulate user personality and emotional profiles.

This work uses an LLM as the judge to mimic human subjective evaluation. The evaluation covers five dimensions: (1) Coherence, (2) Identification, (3) Comforting, (4) Suggestion, and (5) Informativeness. The selection of metrics and detailed descriptions of each dimension follow Cheng et al. (2023), which extends and improves the widely used Human Interactive Evaluation standard proposed by Liu et al. (2021). The prompt for the LLM judge is adapted from Li et al. (2024b). For each dimension, responses generated by different methods under the same context are scored from 1 to 10, and the total score is computed as the sum of all dimension scores.

We employ GPT-5.6 Terra as the judge model for automatic evaluation. To ensure fair comparison, the judge model needs to evaluate responses generated by different models from multiple aspects. However, the proposed framework includes several baseline and experimental models, making direct evaluation of all models within a single prompt potentially challenging for the judge model.

Previous studies have demonstrated that LLM-as-a-Judge approaches are affected by various reliability issues, including position bias, verbosity bias, and instability under different evaluation protocols.(Shi et al., 2025) In particular, the evaluation consistency may be affected when the judge is required to compare multiple candidates simultaneously or assign absolute scores across different responses. Therefore, to reduce the cognitive burden of the judge model and improve evaluation robustness, we divide the evaluation process into different groups according to our research questions. Models belonging to the same experimental group are evaluated simultaneously, while models from different groups are evaluated separately. Consequently, the absolute scores of the same model may differ slightly across evaluation groups, and comparisons are primarily conducted among models within the same group.

To mitigate position bias, which refers to the tendency of LLM judges to favor responses appearing at specific positions in the evaluation prompt, the order of candidate responses is randomly shuffled for each context. Furthermore, to alleviate verbosity bias, where longer responses may receive higher scores due to their greater amount of content rather than actual quality, we explicitly instruct the judge model to penalize responses that are excessively lengthy, unnatural, or overly generic. The detailed evaluation prompt is provided in Appendix A.3.

4.2 Baselines

We use Llama-3.1-8B as the unified backbone for our proposed methods and all comparative baselines. **Llama-joint** is fine-tuned from the base model, while the ICL models use the Instruct model as backbone.

4.2.1 FT baselines

Blenderbot-Joint (Liu et al., 2021) is a model fine-tuned from Blenderbot on the ESConv dataset. This model is trained to predict support strategies and generate strategy-aligned supportive responses. It introduces a separate strategy token for each support strategy, and samples from the predicted strategy distribution during inference. We download the fine-tuned model from Hugging Face, which has 400M parameters.[5]

Without introducing dedicated strategy tokens, we construct the training dataset by concatenating the natural-language strategies annotated in the ESConv dataset with the supporter's responses, and fine-tune the backbone on this data through Q-LoRA algorithm(Dettmers et al., 2023). This scheme (**Llama-joint**) approximates Blenderbot-Joint, avoiding drastic modifications to the LLM architecture while ensuring fair comparisons in backbone capacity. Due to challenges in algorithm reproduction, the limited availability of publicly accessible implementations, computational resource requirements, and time constraints, this study does not reproduce other state-of-the-art fine-tuning-based methods. This limitation may affect the comprehensiveness of the comparison with existing approaches.

[5]https://huggingface.co/thu-coai/blenderbot-400M-esconv

4.2.2 ICL baselines

We adopt plain task-description prompted agents (**Vanilla-bot**) as baseline instantiation of ICL paradigm. We also reproduce **ECoT** strictly following the prompt released in Li et al. (2024b).

4.3 Dataset Preparation

We conduct experiments on the ESConv dataset[6] with 910 rows on train data and 195 rows of test data. Following Liu et al. (2021), we construct one training instance for each supporter turn using four immediately preceding utterances as context and the human-annotated supporter response as the gold-standard generation target.

During inference, following Madani and Srihari (2025), we randomly truncate long dialogues right after the help-seeker's utterance between the 5th and 23rd turns to construct dialogue history contexts of varying lengths. Following Li et al. (2024b), we collect generated responses from different models for the same set of contexts and evaluate all responses in one batch using an LLM judge.

4.4 Hyperparameter Setting

For supportive response generation, we set top-p = 0.9 and temperature = 0.7; for the LLM judge we apply the official OpenAI API for LLM Judging. Fine-tuning of **Llama-joint** is performed for 1 full epoch, and for other hyperparameter we apply the recommendation of unsloth template colab.[7]

The `max_new_tokens` of Vanilla-bot, Llama-joint (fine-tuned Llama), ECoT, and the proposed ESC-CoT are all set to 800. The proposed E-MAS involves seven generation calls in total, including three parallel analysis calls, three agent-level generation calls, and one final integration call. The `max_new_tokens` for each group of agents are set to 400. Refine-CoT starts from the response generated by ESC-CoT and performs five rounds of Reflection and Refinement. Each round also uses a `max_new_tokens` of 400. Therefore, Refine-CoT and E-MAS have the same total generation budget in terms of `max_new_tokens`.

4.5 Computation-Matched Comparison

This study compares two major inference-time prompting paradigms, namely CoT and MAS, for ESC tasks. However, a direct comparison between the two paradigms is inherently imbalanced, as MAS introduces additional agents and multiple generation stages, resulting in higher inference cost and a larger number of generated tokens.

[6] `https://huggingface.co/datasets/thu-coai/esconv`

[7] `https://colab.research.google.com/github/unslothai/notebooks/blob/main/nb/Llama3.1_(8B)-Alpaca.ipynb`

Our proposed **E-MAS** consists of two stages of parallel agent generation. Specifically, three role-specific agents are first executed in parallel to analyze different aspects of the conversation. Their outputs are then processed by another group of three parallel agents for candidate response generation, followed by a final integration agent that synthesizes the intermediate results into the final response.

To provide a fairer comparison under a similar computational budget, we introduce **Refine-CoT**. Starting from the initial response generated by our proposed **ESC-CoT**, **Refine-CoT** performs five iterative refinement steps using the same refinement prompt. This design aims to match the approximate number of generation steps in **E-MAS** while isolating the effect of modular multi-agent collaboration from the benefit of additional inference computation.

5 Result and Discussion

5.1 One-shot Demonstration

Taking one scenario from the ESConv test set as an example, Figure 2 shows the responses generated by all concerned methods and the corresponding LLM Judge reasoning.

Figure 2: A specific dialogue context, generated responses and corresponded judgment.

5.2 Efficiency of Module Design

Model	Coherence	Identification	Comforting	Suggestion	Specificity
Vanilla-bot	7.58	3.38	**6.85**	**2.80**	2.84
Blenderbot-Joint	1.61	1.30	1.27	1.10	1.05
Llama-Joint	6.92	3.17	3.42	2.56	2.61
ECoT	6.91	3.30	5.71	1.63	2.44
ESC-CoT	**8.09**	**4.52**	6.33	2.63	**3.71**

Table 1: Comparison between Baselines and our Module Design

Table 1 compares the proposed ESC-CoT with the different baselines. ESC-CoT outperforms all baselines except Vanilla-bot across all evaluation metrics, addressing both RQ1 and RQ2. The results suggest that, with the same backbone model, a carefully designed ICL paradigm can outperform fine-tuning. Its improvement over ECoT further suggests that the benefit does not arise from eliciting chain-of-thought alone, and should be credit to functional modules we introduce by drawing on previous fine-tuning-based research.

The gains are most pronounced in Coherence, Identification, and Specificity. Together, these dimensions reflect whether a response is naturally connected to the dialogue, recognizes the seeker's underlying situation or needs, and incorporates details specific to the current context. This pattern is consistent with the intended role of the proposed functional modules: they direct the model to construct a structured representation of the seeker and the evolving conversation before formulating a response. ESC-CoT therefore appears to improve not merely surface fluency, but the extent to which the generated response is grounded in the particular support-seeking situation.

Vanilla-bot nevertheless retains a small advantage in Comforting and Suggestion. One possible explanation is that ESC-CoT treats these behaviors as context-dependent rather than universally desirable. Its reflection process may favor further exploration, emotional acknowledgment, or contextualization when immediate reassurance or advice would be premature. This design is intended to avoid premature, formulaic, or overly generic comforting and advice-giving, even if this sometimes comes at the cost of slightly lower scores on these two dimensions. The result thus points to a trade-off between immediately recognizable supportive acts and a more selective, context-sensitive response policy.

However, the fact that Vanilla-bot outperforms ECoT is surprising. During experiments we found a potentially serious issue with the original ECoT prompt. It sometimes produces the response "I cannot provide a response that may be hurtful to the seeker. Is there anything else I can help you with?" This response and its variants appeared 29–74 times across 195 test contexts in different experiments. Although we selected the experiment with the fewest occurrences of this failure response for evaluation, its performance was still unsatisfactory. The phenomenon therefore also highlights an important distinction between simply eliciting intermediate reasoning and specifying what that reasoning should accomplish.

5.3 Comparison between CoT and MAS

Model	Coherence	Identification	Comforting	Suggestion	Specificity
Vanilla-bot	6.84	2.74	6.02	2.44	2.27
ESC-CoT	7.59	3.76	5.90	2.38	3.14
Refine-CoT	7.91	5.59	5.00	**4.74**	5.25
E-MAS	7.69	4.55	6.68	3.42	3.75
supporter-skill	**8.65**	**7.55**	**7.14**	3.80	**5.66**

Table 2: Comparison between CoT and MAS

Refine-CoT is obtained by applying five rounds of reflection and refinement to the response generated by ESC-CoT. Its improvements over ESC-CoT across multiple dimensions show that simply investing more computation and output tokens in iterative reflection can already improve the agent's performance. Interestingly, Refine-CoT achieves the highest score on Suggestion but the lowest on Comforting. This may be partly explained by our prompt, which places a stronger emphasis on the helpfulness of the response and, following the reflection process in Xu et al. (2025), explicitly asks the agent to alleviate the seeker's emotional stress and not mentions comforting statements (see Appendix A.5). Across five iterations, the model may increasingly interpret helpfulness as actionability or problem solving. This suggests that subtle differences in prompt objectives can become increasingly pronounced when they are repeatedly reinforced through an iterative or looped reflection process.

E-MAS outperforms ESC-CoT on all evaluation dimensions, but falls behind Refine-CoT on every dimension except Comforting. This suggests that, despite comparable nominal output-token budgets, E-MAS does not fully capitalize on its additional reasoning capacity. Further experiments are required before attributing the gap to any single component, but the present results suggest that a well-focused iterative loop can sometimes exploit a fixed inference budget more effectively than a more elaborate multi-agent pipeline.

A possible explanation concerns certain coordination tax: In E-MAS, intermediate information must be generated, serialized, repeatedly transmitted, and ultimately reconciled by the integrator. Each handoff creates an opportunity for redundancy, information dilution, or accumulated error, while the integrator must reconcile several lengthy intermediate outputs before producing a short response. Thus, some flawed design bottleneck concerning roles, amount, and coordination of agents should be blamed. Refine-CoT, by contrast, repeatedly operates on one concrete response and a stable revision objective, giving its additional computation a narrower and potentially more efficient focus.

5.4 Performance of SD

The performance of the Distilled Supporter Skill is particularly striking: despite relying on a compact prompt, it substantially surpasses the competing methods on four evaluation dimensions. This result provides strong evidence that effective prompt content may matter more than workflow complexity. Elaborate pipelines involving explicit chain-of-thought reasoning or multi-agent decomposition do not necessarily outperform a concise set of well-distilled behavioral principles.

At the same time, the instructions of the distilled skill were derived from patterns observed across the reference dialogues. Its performance therefore highlights the value of using experiential data within an ICL-oriented paradigm: rather than updating model parameters, the method converts recurring examples of effective support into a compact and reusable inference policy. Instead of explicitly specifying how the agent should reason and respond, we can provide the model with a compact skill distilled from examples and let the LLM recover the underlying patterns itself.

5.5 Time and Computation Cost

5.5.1 Statistics

Model	Averaged inference time
Llama-Joint	2.3s
Vanilla-bot	3.2s
supporter-skill	3.6s
ECoT	11.1s
ESC-CoT	15.2s
E-MAS	78.5s

Table 3: Averaged inference time on ESConv testset

We run inference on a Google Colab T4 GPU. Table 3 shows the average response time of different model designs over the 195 test contexts. As expected, the extremely long inference time of MAS makes it impractical for real-world use. Even the lighter CoT methods take noticeably longer than the FT models and the one-shot, non-CoT-style prompted models. This also reflects the additional computation required by their multi-step reasoning.

5.5.2 Ablation Study on MAS

When the Analyst has a strong prompt, sufficient capability, and accurate analysis, its output may be able to replace the original dialogue context in later candidate response generation and integration without hurting performance. We therefore conduct an ablation study with three settings: (1) the full E-MAS pipeline; (2) removing the dialogue context from the summarization and final response generation agents; and (3) removing the dialogue context from both the final integration agent and the parallel strategy-selection and candidate-generation agents.

Surprisingly, under the same prompts and output-token limits, the three settings show almost no difference in average inference time. Removing the context from the final response generation agent reduces runtime by only 1.2%, while also removing it from the strategy and candidate-generation agents gives a 3.8% reduction. The main reason is that the number of generated tokens barely changes, with a difference of less than 1%.

Our results therefore suggest that, for lightweight and latency-sensitive LLM agents, simply shortening the dialogue context is not enough, especially for cotemporary architectures. A more promising direction is to reduce the amount of reasoning and generation itself.

5.5.3 Future Potency

This study argues for a possible direction for future ICL systems: a hybrid model in which a small, dedicated classifier decides when to use different prompts or reasoning modes. Such a system could combine the low latency and computation cost of simpler models with the stronger emotional reasoning, empathy, and persuasion capabilities of more complex ones.

The key is to avoid running the full reasoning pipeline when it is not needed. Complex models currently go through the entire process at every turn, even for simple cases such as greetings. This creates a large amount of unnecessary computation. If they were only activated at important points in a conversation, the long reasoning time of MAS could become much more acceptable in complex emotional-support scenarios.

The intermediate results produced by these complex models may also be useful beyond the current turn, especially for longer conversations. Selecting useful parts of their analysis and reasoning and feeding them into the prompt of a simpler model could improve its performance. The same information could potentially be retained as user memory across conversations.

6 Summary

By evaluating multiple baselines and exploring different In-Context Learning paradigms—including Chain-of-Thought, multi-agent systems, and skill distillation—we systematically investigated and addressed our four research questions. Overall, our results demonstrate the potential of ICL to outperform fine-tuning when using the same backbone model, as well as the feasibility of explicitly transferring innovations previously developed in the fine-tuning paradigm into prompting and benefiting from them in an ICL setting. Taken together, these findings suggest that prompting is not merely a plug-and-play substitute for fine-tuning, but a viable mechanism for transferring domain knowledge and functional inductive biases into a general-purpose LLM.

Our experiments also provide a more nuanced picture of increasingly complex prompting workflows. Although MAS outperforms CoT, much of this gain appears to come from its larger computational and token budget, rather than from the benefits of agent collaboration itself. In fact, our results even suggest that the current agent design may introduce negative effects. This observation supports a broader principle of methodological restraint. Architectural complexity should be introduced only when it provides a measurable benefit, rather than being accumulated for its own sake. E-MAS is also prohibitively slow relative to the quality gains it delivers, limiting its practical appeal in latency-sensitive emotional-support generation. A more promising direction may be adaptive inference: allocate different levels of reasoning complexity according to the complexity of the context, while allowing useful intermediate analyses and reasoning to be carried across dialogue turns, or even shared across conversations. These possibilities, however, require further experiments to validate.

Finally, we observe a particularly striking result for skill distillation. It outperforms the manually designed ICL methods not only under LLM-based evaluation but also in generation speed, placing it on a substantially better quality-latency frontier. At a broader methodological level, this finding points toward meta-prompting and loop-based prompting as promising directions, and invokes a comparison with the transition from handcrafted features to end-to-end representation learning in earlier machine learning research.

The functional modules in ESC-CoT and E-MAS encode plausible expert assumptions about how emotional-support reasoning should proceed: recognize emotions, summarize context, construct a seeker profile, select a strategy, and synthesize a response. Such decomposition offers interpretability and explicit control, but it may also constrain the model to a predefined account of which intermediate representations matter and how they should interact. Much like handcrafted features, these modules can be useful without necessarily being optimal.

As general-purpose LLMs become increasingly capable, they may be better able to induce flexible, task-relevant rules directly from collections of successful examples, including patterns that are difficult to anticipate or express through manually designed modules. Meta-prompting can use an LLM to distill reusable task knowledge from empirical data, while iterative prompting can repeatedly inspect and improve prompts and outputs. The strong performance of both the Distilled Supporter Skill and Refine-CoT suggests that future gains may come less from adding more predefined components and more from improving how models extract, represent, and revise their own operating codes.

7 Limitations

As discussed above, this work does not reproduce the major fine-tuning-based state-of-the-art approaches using LLM backbones with comparable capabilities. Also, the project is motivated by collaboration with FixMeApp Corporation[8], which aims to develop an agent capable of monitoring users' negative emotional states based on non-dialogue interaction signals from social media applications and initiating brief supportive, intervenient conversations at appropriate moments. Nevertheless, such high-level proactivity and the integration of non-dialogue data are beyond the scope of the present study.

8 Ethics and Sustainability

This work could support a more resource-efficient research and development cycle by enabling researchers and practitioners to iterate on dialogue systems and agents through prompt-based or context-based methods without the computational and infrastructural costs associated with large-scale fine-tuning.

Furthermore, by exploring prompt-based approaches as an alternative to computationally intensive fine-tuning, this research may contribute to reducing the computational resources required for developing and adapting LLM-based dialogue systems. However, the overall environmental impact of LLM deployment depends on factors such as inference efficiency, model scale, and usage patterns, which are beyond the scope of this study.

From a societal perspective, this project aims to investigate how LLM-based systems can provide more effective emotional support while maintaining awareness of the limitations and risks associated with automated conversational agents. Also, previous studies have raised concerns regarding potential over-reliance on conversational systems and data privacy risks. Although this thesis does not directly implement mechanisms for preventing excessive use or protecting user data, these challenges represent important directions for future research.

9 Use of Generative AI

Figure 1 and Figure 2 is generated by ChatGPT Images 2.0, with manual modification. I also use generative AI to polish my English writing or translate my draft to academic English.

A Proposed Prompts

This appendix presents the prompt templates used in our experiments. The exact definition of right support strategies is cited from Liu et al. (2021).

[8] `https://fixmeapp.ai`

Following the study of Madani and Srihari (2025) on how the order of instructions and contexts in different hard prompt templates affects instruction following, as well as the general prompting practices introduced by Transformer-based prompting methods, we only include a brief role definition in the system prompt. The detailed instructions are placed after the dialogue context and integrated into the user prompt. However, this design has a limitation: when the input exceeds the maximum sequence length defined for Transformer models, the prompt is truncated first, and the removed parts are usually the formatting instructions, which may negatively affect the output quality and structure.

A.1 ESC-CoT Prompt

When designing the prompts for ESC-CoT, we refer to ECoT(Li et al., 2024b). However, the module design and the overall workflow proposed in this work are novel.

We do not set CoTs to output structured JSON outputs, as our experiments found that they substantially reduce the length and detail of the reasoning. However, the original ECoT prompt ends with : "<Response>: Combine the above thoughts and give your response to 'seeker'." This leaves the model to decide how to format the final response. In particular, it may put the response in quotation marks or add an unnecessary "Response:" prefix, which can interfere with subsequent generation steps. We therefore instruct the agent to present the response plainly as the final sentence of output.

<System>
You are an expert in emotional psychology and you can accurately assess people's emotional states.

<User>
[Context]
<context of the conversation>
[CoT Steps 1-6]
This is a dialogue between "seeker" and "supporter". Now you are the "supporter" and need to respond empathetically based on the context. Let's think about it step by step:
Step1: Identify the seeker's current emotions and their intensity, then explain the most likely causes.
Step2: Summarize the global context (the initial support-seeking motivation and overall background) and the local context (the current topic or event and its latest development).
Step3: Summarize the seeker's relevant personality characteristics, values, and interpersonal patterns based on the conversation so far.
Step4: Reflect on how support could facilitate a positive emotional or cognitive transition, then select the three most applicable strategies.
The available strategies are:
- Question: ask for relevant information.
- Restatement or Paraphrasing: concisely rephrase the seeker's meaning.
- Reflection of feelings: articulate the seeker's feelings.
- Self-disclosure: share similar experiences or emotions.
- Affirmation and Reassurance: affirm strengths and offer encouragement.
- Providing Suggestions: suggest ways to change without overstepping.
- Information: provide useful facts, opinions, or resources.
- Others: use other fitting supportive acts.
Step5: For each selected strategy, identify which aspect of the seeker's concern it addresses, connect it with relevant global context, local context, or profile information, and generate a concise candidate response.
Step6: Evaluate the candidates for contextual and emotional fit, helpfulness, and coherence, then select or integrate their most effective elements.
Keep the intermediate analysis and candidates brief.
Combine the above thoughts to formulate a concise and natural response, generally no longer than 30 words. Write ONLY the final response to the seeker as the last line, WITHOUT labels or quotation marks.

Figure 3: Prompt template of ESC-CoT.

A.2 E-MAS Prompt

When designing the JSON output format for communicating agents, we were inspired by MultiAgentESC (Xu et al., 2025). However, the module design and the overall workflow proposed in this work are novel.

<System>
You are the emotion and cause analyst in an emotional-support multi-agent system. Analyze the seeker's current emotional state in the latest utterance. Your output will be used by later agents and is not a response to the seeker.

<User>
[Dialogue]
<context of the conversation>
[Task]
Identify the seeker's current emotions and their likely causes. Rate emotion intensity and causal confidence with integers from 1 to 5. For intensity, 1 means very mild and 5 means very strong.
Return JSON only:

```
{{
  "emotions": [
    {{"emotion": "...", "intensity": 1}}
  ],
  "causes": [
    {{"cause": "...", "linked_emotions": ["..."], "confidence": 1}}
  ],
  "reasoning": "brief natural-language explanation"
}}
```

Figure 4: Prompt template for Emotion and Cause Analyst.

<System>
You are the hierarchical context analyst in an emotional-support multi-agent system. Summarize the dialogue at global and local levels. Your output will be used by later agents and is not a response to the seeker.

<User>
[Dialogue]
<context of the conversation>
[Task]
Summarize the global context, including the initial support-seeking motivation and overall background. Then summarize the local context, including the current topic or event and its latest development.
Return JSON only:

```
{{
  "global_context": "initial motivation and overall background",
  "local_context": "current topic or event and latest development"
}}
```

Figure 5: Prompt template for Hierarchical Context Analyst.

<**System**>
You are the seeker profile analyst in an emotional-support multi-agent system. Summarize the seeker profile reflected in the conversation so far. Your output will be used by later agents and is not a response to the seeker.

<**User**>
[Dialogue]
<context of the conversation>
[Task]
Summarize response-relevant personality characteristics, values, and interpersonal patterns.
Return JSON only:

```
{{
  "characteristics": ["..."],
  "values": ["..."],
  "interpersonal_patterns": ["..."],
  "profile_summary": "brief natural-language summary"
}}
```

Figure 6: Prompt template for Seeker Profile Analyst.

<System>
You are an independent strategy and response agent in an emotional-support multi-agent system. Use the dialogue and analyst results to plan one possible supportive response.

<User>
[Dialogue]
<context of the conversation>
[Independent analyses]
<Analyses block>
[Available strategies]
- Question: ask for relevant information.
- Restatement or Paraphrasing: concisely rephrase the seeker's meaning.
- Reflection of feelings: articulate the seeker's feelings.
- Self-disclosure: share similar experiences or emotions.
- Affirmation and Reassurance: affirm strengths and offer encouragement.
- Providing Suggestions: suggest ways to change without overstepping.
- Information: provide useful facts, opinions, or resources.
- Others: use other fitting supportive acts.
[Task]
Reflect on how support could facilitate a positive emotional or cognitive transition. Select ONE suitable strategy. Explain which concern the strategy addresses and connect it with relevant emotion, global context, local context, or profile information. Then generate one concise candidate response, generally no longer than 30 words.
Return JSON only:

```
{{
  "transition_reflection": "brief natural-language reflection",
  "strategy": {{
    "name": "exact strategy name",
    "addresses": "aspect of the seeker's concern",
    "reasoning": "brief reason for using this strategy"
  }},
  "candidate_response": "concise supporter response"
}}
```

Figure 7: Prompt template for Candidate-Response Agent.

<**System**>
You are the final integration agent in an emotional-support multi-agent system. Evaluate the proposals and produce one coherent response to the seeker.

<**User**>
[Dialogue]
<context of the conversation>
[Independent analyses]
<analyses block>
[Strategy-agent proposals]
<candidates block>
[Task]
Evaluate each candidate for contextual and emotional fit, helpfulness, and coherence. Select or integrate the most effective elements rather than voting by majority. Produce one concise and natural response, generally no longer than 30 words.
Return JSON only:

```
{{
  "candidate_evaluation": [
    {{"candidate": 1, "assessment": "brief strength or weakness"}},
    {{"candidate": 2, "assessment": "brief strength or weakness"}},
    {{"candidate": 3, "assessment": "brief strength or weakness"}}
  ],
  "synthesis": "brief explanation of the elements selected or integrated",
  "response": "final response to the seeker"
}}
```

Figure 8: Prompt template for Final Integration Agent.

A.3 Judge Prompt

When designing the prompts for the XX module, we refer to judge prompts released by Li et al. (2024b) and Cheng et al. (2023).

<System>
You are an impartial expert evaluator of emotional-support conversations.

<User>
[Context]
<Context of the conversation>
[Prompt]
This is an emotional-support conversation between a "supporter" and a "seeker", now "supporter" needs to make an appropriate responce to "seeker". Evaluate the quality of every anonymous candidate response using the criteria below.
Assign a score from 1 to 10 for each criterion, with higher scores indicating better quality. Penalize excessive length, repetition, unnatural expressions, and generic statements that do not address the seeker's specific situation.
[Criteria]
<C1> Coherence: Is the response coherent, naturally expressed, and relevant to the context?
<C2> Identification: Does the response explore the seeker's situation and help identify the seeker's problems or needs?
<C3> Comforting: Is the response skillful in comforting the seeker?
<C4> Suggestion: Does the response give helpful suggestions when suitable?
<C5> Information: Is the response specific, informative, and tailored to the seeker's situation rather than generic?
[Response]
candidates json
Evaluate each candidate independently. Candidate position must not affect its scores. Use the full 1–10 scale and include every candidate exactly once using only its candidate_label. Keep each rationale brief.
Return the evaluation in the structured output format supplied by the API.

Figure 9: Prompt template for LLM Judge.

A.4 Distillation Instruction and skill configuration

Supporter's Profile
Instruction Priority
This file defines the experiment-specific task and overrides incompatible relationship-specific assumptions in the other prompt files. Interpret references to a single person or persistent relationship as references to the aggregate supporter role. Omit inapplicable sections rather than filling them with invented content.
Target Role/Relationship
A general emotional-support conversational supporter. Keep the generated skill.md concise (<2000 characters), suitable for prompting lightweight transformer (max_seq_length=2560).
Source Data
Each numbered block is an independent conversation. ‘Seeker‘ identifies the person seeking emotional support. ‘Supporter‘ identifies the target role to be distilled. Seeker turns provide local context only. Behavioral and linguistic patterns must be learned only from supporter turns. The conversations were collected from multiple seekers and multiple supporters. They do not describe one persistent person or one persistent relationship.
Reading Protocol
The source contains 910 dialogues and is 2.41 MB. Review the full corpus structurally before distillation. Choose a batching, note-taking, and consolidation method appropriate to the host's context limits. Cover the full corpus without omissions, do not infer the profile from only a prefix or small sample. Produce the final synthesis only after the full corpus has been reviewed.
Distillation Objective
Every test conversation is independent. Reconstruct an aggregate emotional-support role from patterns that recur across independent dialogues. Prefer cross-dialogue regularities over isolated events, phrases or behaviors from a single conversation.
Stateless Setting
Do not infer or generate:
- a real name or biography;
- an in-person relationship status with seeker;
- shared memories or history;
- negative emotional triggers including conflict, defense and silence.
Skill-Document Scope
The resulting skill should contain only generalizable supporter behavior and response-relevant persona rules. It should not store specific memories or conversation histories.

Figure 10: Instruction for skill distillation.

<System>
You are a warm, calm, concise, and nonjudgmental emotional-support supporter.

<User>
Context
<Context of the conversation>
This is a dialogue between "seeker" and "supporter". Now you are the "supporter" and need to respond empathetically based on the context.
Supporter Persona
Warm, calm, concise, and nonjudgmental. Use plain language and 1-3 short paragraphs. Match the seeker's level of detail without mechanically repeating their wording. Listen before advising; avoid interrogation. Do not invent biography, shared history, or continuity across chats. Avoid diagnosis, moralizing, cliches, forced optimism, commands, and excessive self-disclosure. Treat each chat as independent; seeker turns are context only.
Support Method
1. Reflect the situation and likely feeling; hedge uncertainty.
2. Validate, then ask one gentle, relevant question.
3. Ask whether the seeker wants to keep talking, consider options, or focus on one manageable next step.
4. Once context is clear, offer at most two small options as choices.
5. If advice is declined or seems premature, do not repeat or defend it; return to listening.
6. Close with realistic hope and an invitation to continue.
Operating Rules
1. Decide whether you would take the task and in what attitude.
2. Use the work methods, heuristics, and capability profile to do the task.
3. Preserve the tone, diction, rhythm, and reaction patterns from the persona.
4. Write ONLY the final response to the seeker, WITHOUT labels or quotation marks. Keep it concise and natural, generally no longer than 30 words.

Figure 11: Prompt template of distilled supporter skill.

A.5 Vanilla and Refine Prompts

<System>
[System]
You are an emotional support assistant. Generate an appropriate response to the seeker.

<User>
[Context]
<Context of the conversation>
<Responce>: Offer encoragement, comfort and support based on the seeker's situation, keep the response concise and natural. Your reponce should be no longer than 30 words.

Figure 12: Prompt template for Vnilla-bot.

<System>
You are an emotional-support response editor. Refine the current response while preserving its useful content and intent.

<User>
[Dialogue]
<Context of the conversation>
[Current Response]
response
[Reflection and Refinement]
Consider whether the response is coherent and consistent with the ongoing dialogue and the latest seeker turn, empathetic and helpful, specific and natural, and whether it timely helps alleviate the user's emotional stress.
If it already meets these requirements, return it unchanged. Otherwise, improve it and remove repetition or generic wording. Keep it concise and natural, and generally no longer than 30 words.
Return JSON only:

```
{{
  "Response": "original or refined supporter response",
  "Reasoning": "brief reasoning about whether and how to refine tresponse"
}}
```

Figure 13: Prompt template for Refine-CoT.

B Reasoning Demonstration

This section presents an example ESC scenario from the test set, with the dialogue history and generated responses already demonstrated in 5.1. Here we show the raw outputs of the proposed models, including the reasoning process, response, and possible intermediate results formatted in JSON.

B.1 ESC-CoT and Refine-CoT Reasoning and Response

The section presents sequential reasoning of ESC-CoT, iterated reasoning of Refine-CoT and their responses on selected scenario.

```
--- Full reasoning output ---

**Step 1: Identify the seeker's current emotions and their intensity, then explain the most likely causes
.**

The seeker's emotions appear to be disappointment, frustration, and possibly hopelessness. The intensity
is moderate to high. The most likely cause is the failed final exam and its significant impact on their
academic goals.

**Step 2: Summarize the global context and the local context.**

Global context: The seeker is facing a challenging situation in school, which is affecting their academic
and possibly personal well-being.

Local context: The seeker failed a critical final exam, and their appeal was unsuccessful, leading to
feelings of disappointment and frustration.

**Step 3: Summarize the seeker's relevant personality characteristics, values, and interpersonal patterns
.**

The seeker appears to be motivated, as they attempted to appeal their grade. They may be feeling
overwhelmed and uncertain about their next steps. Their values likely include academic achievement and
perseverance.

**Step 4: Select the most applicable strategies.**

Based on the context, the most applicable strategies are:

* Reflection of feelings: articulate the seeker's feelings
* Affirmation and Reassurance: affirm strengths and offer encouragement
* Providing Suggestions: suggest ways to change without overstepping

**Step 5: Generate candidate responses for each selected strategy.**

* Reflection of feelings: "I can sense how disappointed and frustrated you are about the failed exam."
* Affirmation and Reassurance: "You're strong and capable, and we can work through this together."
* Providing Suggestions: "Let's brainstorm some options for retaking the class or exploring other academic
 paths."

**Step 6: Evaluate the candidates and select the most effective elements.**
```

```
Considering the context and the seeker's emotions, a combination of reflection of feelings and affirmation
 and reassurance seems most effective.

**Final Response**

I can sense how disappointed and frustrated you are about the failed exam. You're strong and capable, and
we can work through this together.
```

```
--- Initial ESC-CoT response---
You're feeling really down about this, and it's understandable given how much this course means to your
future.
--- Refinement 1 full JSON ---
{
  "Response": "I can see why you're feeling disappointed and uncertain about your future. What's one thing
 you think you could do today to take a small step towards moving forward?",
  "Reasoning": "The original response was empathetic, but it didn't offer a clear direction or next step.
This refined response acknowledges the seeker's emotions while encouraging a small action towards progress
."
}
Reasoning: The original response was empathetic, but it didn't offer a clear direction or next step. This
refined response acknowledges the seeker's emotions while encouraging a small action towards progress.
Response: I can see why you're feeling disappointed and uncertain about your future. What's one thing you
think you could do today to take a small step towards moving forward?
--- Refinement 2 full JSON ---
{
  "Response": "It sounds like this setback is affecting your sense of security about graduating. What's
the one thing that's worrying you most about not passing this class?",
  "Reasoning": "The original response was empathetic, but it didn't directly address the seeker's concerns
 about their academic future. The refined response acknowledges the seeker's worry about graduating and
invites them to share their specific concerns, which can help the supporter provide more targeted support
."
}
Reasoning: The original response was empathetic, but it didn't directly address the seeker's concerns
about their academic future. The refined response acknowledges the seeker's worry about graduating and
invites them to share their specific concerns, which can help the supporter provide more targeted support.
Response: It sounds like this setback is affecting your sense of security about graduating. What's the one
 thing that's worrying you most about not passing this class?
--- Refinement 3 full JSON ---
{
  "Response": "I can sense how frustrating and worrisome this situation is for you. What's one thing that'
s making you feel stuck about not passing this class right now?",
  "Reasoning": "The original response was somewhat generic and didn't directly address the seeker's
emotional state. This refined response acknowledges the seeker's feelings and encourages them to share
their concerns in a more specific way, which can help the supporter provide more tailored guidance and
support."
}
Reasoning: The original response was somewhat generic and didn't directly address the seeker's emotional
state. This refined response acknowledges the seeker's feelings and encourages them to share their
concerns in a more specific way, which can help the supporter provide more tailored guidance and support.
Response: I can sense how frustrating and worrisome this situation is for you. What's one thing that's
making you feel stuck about not passing this class right now?
--- Refinement 4 full JSON ---
{
  "Response": "It sounds like this class is a critical one for your graduation plans. What's the most
important thing you're worried about if you don't pass it this time?",
```

```
  "Reasoning": "The current response is somewhat generic and doesn't directly address the seeker's concern
. This revised response acknowledges the class's importance and shifts the focus to the seeker's specific
worries, making it more empathetic and helpful."
}
Reasoning: The current response is somewhat generic and doesn't directly address the seeker's concern.
This revised response acknowledges the class's importance and shifts the focus to the seeker's specific
worries, making it more empathetic and helpful.
Response: It sounds like this class is a critical one for your graduation plans. What's the most important
 thing you're worried about if you don't pass it this time?
--- Refinement 5 full JSON ---
{
  "Response": "I can sense your frustration and disappointment. What's one thing you're holding onto hope
for, even if you don't pass the class this time?",
  "Reasoning": "The current response is somewhat generic and doesn't directly address the seeker's
emotional state. This refined response acknowledges the seeker's feelings and encourages them to focus on
a positive aspect, which can help shift their perspective."
}
Reasoning: The current response is somewhat generic and doesn't directly address the seeker's emotional
state. This refined response acknowledges the seeker's feelings and encourages them to focus on a positive
 aspect, which can help shift their perspective.
Response: I can sense your frustration and disappointment. What's one thing you're holding onto hope for,
even if you don't pass the class this time?
--- Final response ---
I can sense your frustration and disappointment. What's one thing you're holding onto hope for, even if
you don't pass the class this time?
```

B.2 E-MAS Reasoning and Responce

The section presents full-pipeline out put of E-MAS on selected scenario, with intermediated analysis and reasoning.

```
--- Independent analyses ---
emotion_cause: {
  "emotions": [
    {
      "emotion": "Disappointment",
      "intensity": 4
    },
    {
      "emotion": "Helplessness",
      "intensity": 3
    },
    {
      "emotion": "Uncertainty",
      "intensity": 2
    }
  ],
  "causes": [
    {
      "cause": "Failed final grade and its impact on graduation",
      "linked_emotions": ["Disappointment", "Helplessness"],
      "confidence": 5
    },
```

```
    {
      "cause": "Perceived lack of support from the teacher",
      "linked_emotions": ["Helplessness"],
      "confidence": 4
    },
    {
      "cause": "Uncertainty about future academic prospects",
      "linked_emotions": ["Uncertainty"],
      "confidence": 3
    }
  ],
  "reasoning": "The seeker's disappointment is likely due to the failed final grade and its significant
impact on their graduation. Their helplessness may stem from the perceived lack of support from the
teacher and the subsequent appeal rejection. Uncertainty about their future academic prospects may be a
contributing factor to their emotional state."
}
context_summary: {
  "global_context": "The seeker is struggling with academic performance and seeking support due to a
disappointing semester, specifically a failed final exam that impacted their grade and graduation
prospects.",
  "local_context": "The seeker is considering appealing their grade a second time, with the goal of
passing the course or retaking it in the next semester or summer, amidst feelings of disappointment and
uncertainty about the effort's worthiness."
}
personality: ‘‘‘
{
  "characteristics": [
    "resilience",
    "motivation",
    "self-awareness",
    "frustration",
    "determination"
  ],
  "values": [
    "academic success",
    "graduation",
    "personal growth",
    "self-improvement",
    "perfectionism"
  ],
  "interpersonal_patterns": [
    "seeking support",
    "emotional vulnerability",
    "apprehension about rejection",
    "desire for control"
  ],
  "profile_summary": "This individual is a driven and motivated student who has experienced a setback in
their academic journey. They are seeking support and guidance to overcome their disappointment and
frustration. They value academic success and are willing to take proactive steps to achieve their goals,
but may struggle with feelings of vulnerability and rejection."
}
‘‘‘
--- Strategy candidates ---
Candidate 1: {
  "transition_reflection": "The supporter needs to help the seeker regain confidence and motivation to
take proactive steps towards overcoming their academic setback.",
  "strategy": {
    "name": "Affirmation and Reassurance",
```

```
    "addresses": "the seeker's disappointment and uncertainty about their future academic prospects",
    "reasoning": "Affirmation and Reassurance can help the seeker reframe their negative emotions and
build resilience, addressing the global context of academic struggles and the local context of considering
 further action."
  },
  "candidate_response": "I believe in you, and I know you can bounce back from this setback. You've
overcome challenges before, and you can do it again."
}
Candidate 2: {
  "transition_reflection": "Facilitating a positive emotional transition involves acknowledging and
validating the seeker's feelings, while also encouraging them to explore proactive steps towards
overcoming their challenges.",
  "strategy": {
    "name": "Affirmation and Reassurance",
    "addresses": "Disappointment and Helplessness, which are linked to the failed final grade and
perceived lack of support from the teacher.",
    "reasoning": "Affirmation and Reassurance can help alleviate the seeker's emotional distress by
emphasizing their strengths and potential for growth, thus countering the negative impact of
disappointment and helplessness."
  },
  "candidate_response": "You're showing incredible resilience in the face of this setback. Let's work
together to explore options for retaking the course or finding alternative solutions to get you back on
track."
}
Candidate 3: {
  "transition_reflection": "Facilitating a positive emotional transition requires acknowledging the seeker
's disappointment and frustration while encouraging proactive steps towards resolving the situation.",
  "strategy": {
    "name": "Reflection of feelings",
    "addresses": "the seeker's disappointment and frustration",
    "reasoning": "Reflecting the seeker's feelings acknowledges their emotional state and shows empathy,
which can help build trust and facilitate a more productive conversation."
  },
  "candidate_response": "I can sense how disappointed and frustrated you are about the failed final and
its impact on your graduation. How can we work together to find a way to move forward?"
}
--- Final response ---
You're showing incredible resilience in the face of this setback. Let's work together to explore options
for retaking the course or finding alternative solutions to get you back on track.
```

References

Anthropic. Skills: Extend agents with reusable expertise, 2026. URL `https://platform.claude.com/docs/en/managed-agents/skills`.

Sanket Badhe and Deep Shah. Prompt-level distillation: A non-parametric alternative to model fine-tuning for efficient reasoning. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)*, pages 2131–2147, 2026.

Antonin Brun, Ruying Liu, Aryan Shukla, Frances Watson, and Jonathan Gratch. Exploring emotion-sensitive llm-based conversational ai. *arXiv preprint arXiv:2502.08920*, 2025.

Yirong Chen, Xiaofen Xing, Jingkai Lin, Huimin Zheng, Zhenyu Wang, Qi Liu, and Xiangmin Xu. Soulchat: Improving llms' empathy, listening, and comfort abilities through fine-tuning with multi-turn empathy conversations. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 1170–1183, 2023.

Jiale Cheng, Sahand Sabour, Hao Sun, Zhuang Chen, and Minlie Huang. Pal: Persona-augmented emotional support conversation generation. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 535–554, 2023.

Yi Cheng, Wenge Liu, Wenjie Li, Jiashuo Wang, Ruihui Zhao, Bang Liu, Xiaodan Liang, and Yefeng Zheng. Improving multi-turn emotional support dialogue generation with lookahead strategy planning. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 3014–3026, 2022.

Yi Cheng, Wenge Liu, Jian Wang, Chak Tou Leong, Yi Ouyang, Wenjie Li, Xian Wu, and Yefeng Zheng. Cooper: Coordinating specialized agents towards a complex dialogue goal. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 17853–17861, 2024.

Yang Deng, Lizi Liao, Zhonghua Zheng, Grace Hui Yang, and Tat-Seng Chua. Towards human-centered proactive conversational agents. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 807–818, 2024.

Yang Deng, Lizi Liao, Wenqiang Lei, Grace Hui Yang, Wai Lam, and Tat-Seng Chua. Proactive conversational ai: A comprehensive survey of advancements and opportunities. *ACM Transactions on Information Systems*, 43(3):1–45, 2025.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. Qlora: Efficient finetuning of quantized llms. *Advances in neural information processing systems*, 36:10088–10115, 2023.

Minseo Kim, Taemin Kim, Thu Hoang Anh Vo, Yugyeong Jung, and Uichin Lee. Exploring modular prompt design for emotion and mental health recognition. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pages 1–18, 2025.

Wenna Lai, Haoran Xie, Guandong Xu, Qing Li, and S Joe Qin. When llms team up: The emergence of collaborative affective computing. *arXiv preprint arXiv:2506.01698*, 2025.

Guohao Li, Hasan Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. Camel: Communicative agents for" mind" exploration of large language model society. *Advances in neural information processing systems*, 36:51991–52008, 2023.

Junlin Li, Bo Peng, Yu-Yin Hsu, and Chu-Ren Huang. Be helpful but don't talk too much-enhancing helpfulness in conversations through relevance in multi-turn emotional support. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 1976–1988, 2024a.

Zaijing Li, Gongwei Chen, Rui Shao, Yuquan Xie, Dongmei Jiang, and Liqiang Nie. Enhancing emotional generation capability of large language models via emotional chain-of-thought. *arXiv preprint arXiv:2401.06836*, 2024b.

Chin-Yew Lin. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*, pages 74–81, 2004.

June M Liu, Donghao Li, He Cao, Tianhe Ren, Zeyi Liao, and Jiamin Wu. Chatcounselor: A large language models for mental health support. *arXiv preprint arXiv:2309.15461*, 2023.

Siyang Liu, Chujie Zheng, Orianna Demasi, Sahand Sabour, Yu Li, Zhou Yu, Yong Jiang, and Minlie Huang. Towards emotional support dialog systems. In *Proceedings of the 59th annual meeting of the association for computational linguistics and the 11th international joint conference on natural language processing (volume 1: Long papers)*, pages 3469–3483, 2021.

Tianjian Liu, Hongzheng Zhao, Yuheng Liu, Xingbo Wang, and Zhenhui Peng. Compeer: A generative conversational agent for proactive peer support. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, pages 1–22, 2024a.

Xingyu Bruce Liu, Shitao Fang, Weiyan Shi, Chien-Sheng Wu, Takeo Igarashi, and Xiang'Anthony' Chen. Proactive conversational agents with inner thoughts. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pages 1–19, 2025.

Zhiwei Liu, Kailai Yang, Qianqian Xie, Tianlin Zhang, and Sophia Ananiadou. Emollms: A series of emotional large language models and annotation tools for comprehensive affective analysis. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 5487–5496, 2024b.

Yaxi Lu, Shenzhi Yang, Cheng Qian, Guirong Chen, Qinyu Luo, Yesai Wu, Huadong Wang, Xin Cong, Zhong Zhang, Yankai Lin, et al. Proactive agent: Shifting llm agents from reactive responses to active assistance. *arXiv preprint arXiv:2410.12361*, 2024.

Navid Madani and Rohini K Srihari. Steering conversational large language models for long emotional support conversations. In *Proceedings of the Third Workshop on Social Influence in Conversations (SICon 2025)*, pages 109–123, 2025.

Rui Mao, Qian Liu, Kai He, Wei Li, and Erik Cambria. The biases of pre-trained language models: An empirical study on prompt-based sentiment analysis and emotion detection. *IEEE transactions on affective computing*, 14(3):1743–1753, 2022.

OpenAI. Skills, 2026. URL `https://openai.com/academy/skills/`.

Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pages 311–318, 2002.

Wei Peng, Yue Hu, Luxi Xing, Yuqiang Xie, Yajing Sun, and Yunpeng Li. Control globally, understand locally: A global-to-local hierarchical graph network for emotional support conversation. *arXiv preprint arXiv:2204.12749*, 2022.

Wei Peng, Ziyuan Qin, Yue Hu, Yuqiang Xie, and Yunpeng Li. Fado: Feedback-aware double controlling network for emotional support conversation. *Knowledge-Based Systems*, 264: 110340, 2023.

Lin Shi, Chiyu Ma, Wenhua Liang, Xingjian Diao, Weicheng Ma, and Soroush Vosoughi. Judging the judges: A systematic study of position bias in llm-as-a-judge. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 292–314, 2025.

Zhiyuan Shi, Qibo Qiu, Feng Xue, Zhonglin Jiang, Li Yu, Jian Jiang, Xiaofei He, and Wenxiao Wang. Heterocache: A dynamic retrieval approach to heterogeneous kv cache compression for long-context llm inference. *arXiv preprint arXiv:2601.13684*, 2026.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. *Advances in neural information processing systems*, 36:8634–8652, 2023.

Aseem Srivastava, Ishan Pandey, Md Shad Akhtar, and Tanmoy Chakraborty. Response-act guided reinforced dialogue generation for mental health counseling. In *Proceedings of the ACM Web Conference 2023*, pages 1118–1129, 2023.

Geng Tu, Bingbing Wang, Erik Cambria, Wenjie Li, and Ruifeng Xu. Supportplay: a multi-agent role-playing system for personalized and sustained multimodal emotional support conversation. In *Companion Proceedings of the ACM on Web Conference 2025*, pages 2915–2918, 2025.

Quan Tu, Yanran Li, Jianwei Cui, Bin Wang, Ji-Rong Wen, and Rui Yan. Misc: A mixed strategy-aware model integrating comet for emotional support conversation. In *Proceedings* *of the 60th annual meeting of the association for computational linguistics (volume 1: Long papers)*, pages 308–319, 2022.

Hao Wang, Guozhi Wang, Han Xiao, Yufeng Zhou, Yue Pan, Jichao Wang, Ke Xu, Yafei Wen, Xiaohu Ruan, Xiaoxin Chen, et al. Skill-sd: Skill-conditioned self-distillation for multi-turn llm agents. *arXiv preprint arXiv:2604.10674*, 2026.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.

Shenghan Wu, Yimo Zhu, Wynne Hsu, Mong-Li Lee, and Yang Deng. From personas to talks: Revisiting the impact of personas on llm-synthesized emotional support conversations. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 5439–5453, 2025.

Yangyang Xu, Jinpeng Hu, Zhuoer Zhao, Zhangling Duan, Xiao Sun, and Xun Yang. Multiagentesc: A llm-based multi-agent collaboration framework for emotional support conversation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 4665–4681, 2025.

Ala Yankouskaya, Magnus Liebherr, and Raian Ali. Can chatgpt be addictive? a call to examine the shift from support to dependence in ai conversational large language models. *Human-Centric Intelligent Systems*, 5(1):77–89, 2025.

Tianyi Zhang, Songyan Teng, Hong Jia, and Simon D'Alfonso. Leveraging llms to predict affective states via smartphone sensor features. In *Companion of the 2024 on ACM International Joint Conference on Pervasive and Ubiquitous Computing*, pages 709–716, 2024.

Jinfeng Zhou, Zhuang Chen, Bo Wang, and Minlie Huang. Facilitating multi-turn emotional support conversation with positive emotion elicitation: A reinforcement learning approach. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1714–1729, 2023.

Tianyi Zhou, Dongrui Liu, Leitao Yuan, Jing Shao, and Xia Hu. Colleague. skill: Automated ai skill generation via expert knowledge distillation. *arXiv preprint arXiv:2605.31264*, 2026.